…… :QPCODE Reg.No :……

Name :……

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) THIRD SEMESTER EXAMINATION

MG3DSCPHY201 Essential Mathematics for Physics

Time:- 2 Hour **Marks:- 70**

Part A

Short Answer Type Questions

Answer any 10 questions

Each question carries 3 marks

$(10 \times 3 = 30)$

1. Define the scalar(Dot) product and explain the dot product in the context of work done by a force. [U][CO:1]

2. Explain the physical concept of vector(cross) product in the context rotational motion in physics? [U][CO:1]

3. Demonstrate the orthogonality(orthonormality) of the unit basis vectors in a right-handed Cartesian coordinate system. [U][CO:1]

4. Define the inverse of a matrix. [An][CO:2]

5. What do you mean a Hermitian matrix and give any two properties. [U][CO:3]

6. If $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, find its eigenvalues and eigenvectors. [A][CO:7]

7. Explain the physical interpretation of divergence for the following cases

   (a) $\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$

   (b) $\nabla \cdot \mathbf{B} = 0$

   Where $\boldsymbol{E}$, $\boldsymbol{B}$ are the electric and magnetic fields respectively. [An][CO:4]

8. Explain the concept of the gradient of a scalar field. Illustrate its significance by discussing the relationship between a potential field and the corresponding force field. [An][CO:4]

9. Give the expression for the $\nabla \boldsymbol{Q}\left(r, \theta, \phi\right)$ in spherical polar coordinate system. [K][CO:5]

10. What is the value of the line integral of a conservative field over a closed loop, and why? [An][CO:6]

11. Can area be expressed as a vector quantity? If so, what determines its direction? If not, explain why? [U][CO:6]

12. State and explain divergence Theorem (Gauss's Theorem) [K][CO:6]

Part B

Short Essay Type Questions

Answer any 8 questions

Each question carries 5 marks

$(8 \times 5 = 40)$

13. Let the vectors $\boldsymbol{a} = (1,1,0), \quad \boldsymbol{b} = (0,1,1), \quad \boldsymbol{c} = (1,0,1)$. Find the reciprocal vectors $\boldsymbol{a}', \boldsymbol{b}', \boldsymbol{c}'$ using the definition: $\boldsymbol{a}' = \dfrac{\boldsymbol{b} \times \boldsymbol{c}}{\boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c})}, \quad \boldsymbol{b}' = \dfrac{\boldsymbol{c} \times \boldsymbol{a}}{\boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c})}, \quad \boldsymbol{c}' = \dfrac{\boldsymbol{a} \times \boldsymbol{b}}{\boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c})}$.

[A][CO:1]

14. A force $\boldsymbol{F}$ acts on a particle at position vector $\boldsymbol{r}$. Explain how the scalar and vector products are used to compute work done and torque, respectively. [A][CO:7]

15. What is a diagonal matrix? How is it related to square matrices. Why are diagonal matrices important in physical applications? [An][CO:2]

16. Solve the following system of equations using the Gauss elimination method:

$$\begin{cases} x + y + z = 6 \\ 2y + 5z = -4 \\ 2x + 5y - z = 27 \end{cases}$$

[A][CO:2]

17. Check whether the following matrix is orthogonal:

$$Q = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

[A][CO:3]

18. Write the expressions for the gradient of a scalar field, and the divergence and curl of a vector field in Cartesian coordinates. Briefly explain their physical interpretations [U,An][CO:4]

19. A scalar field is given by $\phi = x^2 + y^2$. Find the gradient at point (1,-2), explain the result. [An][CO:4]

20. Show that $\nabla \times (\mathbf{r} \times \mathbf{F})$ is generally not zero, and explain what this means physically for the force field [E][CO:7]

21. What does it mean for a force field to be path-independent, and what is the mathematical condition for this property? [A][CO:6]

22. Let $\mathbf{F} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$ and let S be the surface of a sphere of radius R centered at the origin. Evaluate the surface integral $\iint_S \mathbf{F} \cdot d\mathbf{A}$ using the divergence theorem. [A][CO:6]